

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 321 — Past Questions & Materials
by Renance

Course: GET 321
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 300

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Paper 1 of 6 — GET 321 - Engineering Mathematics IV

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING,
COLLEGE OF ENGINEERING
SECOND SEMESTER EXAMINATION, 2024/2025 SESSION
COURSE CODE AND TITLE: GET 321 – ENGINEERING MATHEMATICS IV

Time Allowed: 2 Hours

Instruction: Answer all questions

1.(a) State the null and alternative hypothesis and represent them symbolically

(b) A survey was made among some drivers to find out those that are running red lights. From the result of the survey, the proportion of drivers who admitted to running red lights is greater than 0.5. Show this claim in null and alternative hypothesis in symbolic form.

(c). With reference to question 1b above, if $n = 750$ randomly selected adult drivers used for the survey, 56% of those that responded admitted to running red lights, find the value of the test statistics for the claim that the majority of all adult drivers admitted to running red lights.

(d). Explain what you understand by test statistics.

25 Marks

2. (a) The life times (in hours) of a certain type of battery are recorded as follows:

Hours	10 - 20	20 - 30	30 - 40	40 - 50
Frequency	5	8	12	5

From the record, Find:

i. The mean lifetime of the battery ii. The Variance iii. The standard Deviation

2b. A box contains six (6) defective and fourteen (14) non-defective bulbs. Two bulbs are selected without replacement. What is the probability that:

i. Both bulbs are defective ii. Exactly one bulb is defective

15Marks

3. Twenty four (24) factory workers were asked to provide the number of hours they worked in the month of August, 2025, their answers are provided below:

48 40 47 41 50 27 52 39 38 36 25 41 35 36 16 40 36 44 40 39 40 43 40 39

From the information provided by these factory workers:

- Calculate the three quartiles
- Find the inter quartile range
- Find the approximate value of the 71st percentile
- Calculate the percentile rank of 40

15 Marks

4. Lagos State University Teaching Hospital management board released the weight readings of some pregnant women for analysis as:

40 42 44 50 55 and 60, find the following:

- The true population mean
- The true variance of weights
- The true standard deviation of weights
- Find a 90% confidence interval for μ , the true mean weight, (Take $Z_{\frac{\alpha}{2}} = 1.645$)

15 Marks

This is a numerical value that uses sample test or data in order to determine whether to reject H_0

Paper 2 of 6 — GET 321 - 2024/2025 - First Semester - ENGINEERING MATHEMATICS IV

Level 300 • First Semester • PastQuestion

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 FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
 DEPARTMENT OF MECHANICAL ENGINEERING,
 COLLEGE OF ENGINEERING
 SECOND SEMESTER EXAMINATION, 2024/2025 SESSION
 COURSE CODE AND TITLE: GET 321 – ENGINEERING MATHEMATICS IV
 Time Allowed: 2 Hours
 Instruction: Answer all questions

1.(a) State the null and alternative hypothesis and represent them symbolically

(b) A survey was made among some drivers to find out those that are running red lights. From the result of the survey, the proportion of drivers who admitted to running red lights is greater than 0.5. Show this claim in null and alternative hypothesis in symbolic form.

(c). With reference to question 1b above, if $n = 750$ randomly selected adult drivers used for the survey, 56% of those that responded admitted to running red lights, find the value of the test statistics for the claim that the majority of all adult drivers admitted to running red lights.

(d). Explain what you understand by test statistics.

25 Marks

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Hours	10 - 20	20 - 30	30 - 40	40 - 50
Frequency	5	8	12	5

From the record, Find:

i. The mean lifetime of the battery ii. The Variance iii. The standard Deviation

2b. A box contains six (6) defective and fourteen (14) non-defective bulbs. Two bulbs are selected without replacement. What is the probability that:

i. Both bulbs are defective ii. Exactly one bulb is defective

15 Marks

$$\frac{\sum fx}{\sum f}$$

$$Z_F = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$Z_{1-\alpha}$$

$$-Z_{1-\alpha}$$

$$SD = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}}$$

$$= \sqrt{Var}$$

$$P(A \cap B) = P(A) + P(B)$$

$$P(A \cup B) = P(A) + P(B)$$

$$\frac{\sum x}{\sum f}$$

$Q_1 = 6.25^{th}$
 $Q_3 = 12.5^{th}$
 18.75^{th}

3. Twenty four (24) factory workers were asked to provide the number of hours they worked in the month of August, 2025, their answers are provided below:

48 40 47 41 50 27 52 39 38 36 25 41 35 36 16 40 36 44 40 39 40 43 40 39

From the information provided by these factory workers:

- Calculate the three quartiles
- Find the inter quartile range
- Find the approximate value of the 71st percentile
- Calculate the percentile rank of 40

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40 42 44 50 55 and 60, find the following:

- The true population mean
- The true variance of weights
- The true standard deviation of weights
- Find a 90% confidence interval for μ , the true mean weight, (Take $Z_{\frac{\alpha}{2}} = 1.645$)

15 Marks

49
 $40 + 1 + 1 + 1 + 1$
 47
 $41 + 1$
 50
 27
 52
 $39 + 1 + 1$
 38
 $36 + 1 + 1$
 25
 35
 16
 40
 43

$\frac{n(n+1)}{4}$

$44 + 46$

Paper 3 of 6 — GET 321 Engineering Mathematics IV 2019 2020 Second Semester

Level 300 • Second Semester • PastQuestion

A aera RS WENO OU RG LINULL UE fig
om OF MECHANICALENGINEERING _
cond Semester Examination 2019/2! i
ie 020 Session
marseeyn eRe TITLE: GET 321- Engineering Mathematics IV
: ANY FOUR (4) QUESTIONS TIME: 2hrs 30mins

QUESTION (1)

(a) ae geo of a certain coating, designed to inhibit corrosion, are being tested. For each of the six
6 ha Pe ee coated with formulation A and half is coated with formulation B and exposed to salt
ater for s. the corrosion losses as measured for each formulation on each pipe is as shown. .

| Pipe A | 197 161 144 162 185 154

\ " Pipe B | 204 | 182 140 178 183 163

From the information provided, using a 95% two sided confidence interval, find the difference
between

the mean corrosion losses of the two formulations?

(b) In a sample of 50 gears, the average lifetime was fo
engineer reported a confidence interval of (11.09, 1

level of this confidence interval?

ard deviation of 6.83, An

fy the level. What is the

17.5 Marks

und to be 12.68 with a stand:

4.27) but neglected to speci

QUESTION (2)

(a) The manager of a store is thinking of establishing a new billing system for the store's credit
customers.

The manager determines that the new system will be cost effective only if the mean monthly
account is

greater than N70. A sample of 200 monthly accounts is drawn for which the sample mean account
is N68.

The account is normally distributed with a standard deviation of N30. Is there enough evidence at
5%

level of significance to conclude that the new system will be cost effective?

sented by region (i) 14 (ii) 17 (iii) 8 and 19

= the event that the car is.due for

(b) From the Venn diagram below state in words the event repre:

= the event that it will take time to

together (iv) 9 (Vv) 8, 19, 7 and 17 together. Assuming that A maintenance, B = the event that spare parts will be required and C finish the maintenance of the car.

[SS

ess

7

c

17.5 Marks

U

QUESTION (3)

st child. The following was

(a) A sample of sixteen (16) oe ay taken, and each asked the age of their fir: their response: 10, 17, 19,21, 2,3, 11, 18, 22, 35, 3, 19, 3, 5, 17,31.

From the data obtained, find (i) The mean (ii) The variance (iii) The standard deviation (v) Range for fingl year students, the following marks were

In a continuous assessment exercise carried out ip Fypre i 8, 1° 3, 21, 13, 10, 8. 18, 52, 26, 23, 19, 9, 14,21, 18, 16, recorded for thirty G) students; 28, 18, 12, 4, 26, 1 33, 12, 14, 13,28, 12, 14, 12, §, 11. Find ercentile rank of 19 (iii) Find the approxim

(i) The three (3) Quartiles (ii) Calculate the p percentile (iv) Quartile coefficient of skewness

17.5 Marks

ate value of the 95"

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(QUES LIUN (4)

lation dat:

(a) The table below shows the data from independent simple random samples from several population data.

North East 16 | 13 17 13 14 12 |

South South [14] 11 12 14 13

SouthEast 9 Ti 8 7 9

South West [12/8 9 13 12 14 16 7

From the information provided,

(i) State the null and alternative hypothesis (ii) Calculate SSB, SSW and SST (iii) Establish the Variance

between samples and Variance within samples (iv) Decide at 0.05 significance level, if the data provides

sufficient evidence to conclude that a difference exist in the mean of the data in which the samples were

drawn. For $\alpha = 0.05$, assume the critical value $F = 2.47$

; 17.5 Marks

QUESTION (5)

(a) What is the difference b

(b) In a gear manufacturing

each day over a fifty (50

etween a population and a sample size in statistics

company, the data below shows the number of gears that failed quality test for

) day period as recorded by the quality control officer,

1, 3, 4, 2, 6, 5, 3, 2, 3, 6, 2, 1, 1, 4, 5, 2, 5, 6, 3, 1, 3, 2, 4, 1, 3, 2, 3, 4, 2, 5, 3, 1, 1, 1, 2, 4, 5, 1, 2, 3, 6, 2;

2, 5, 1, 2, 5, 2, 5, 4, 4, 5: ' .

From the information provided,

(i) Construct a frequency distribution table

Gi) Find the relative frequencies and the percentage distribution

(iii) Calculate the relative frequency of the gears having three (3) or more failures per a day

(iv) Calculate the frequency of gears having two (2) or five (5) failures per day

(v) Draw a bar graph for the percentage distribution of the data obtained

(c) Differentiate between point estimate and interval estimate

17.5 Marks

QUESTION (6)

(a) An article describes an experiment in which eight (8) steel specimens were submerged in seawater at

various temperatures, and the corrosion rates were measured. The results are presented in the following table

Temp.(C) (x) | 48.3 | 36.5 | 32.2 | 17.9 | 22.4 | 31.9 | 41.3 | 12.6 |

Corrosion (y) | 10.2 | 6.82 | 5.8 | 2.3 | 3.5 | 4.7 | 8.2 | 1.6

(mm) "

(i) From the available data, use the linear least square method to determine the coefficients function $y = a + bx$ that best fit the data. -

(ii) The standard deviation error S- (iii) The error sum of squares SSE (iv) The total sum of Squares SST

(vi) Predict the corrosion rate for steel submerged in seawater at a temperature of 20°C

"a" and "b" into the

17.5 Marks .

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Paper 4 of 6 — GET 321 MECHANICAL ENGINEERING 2024 past question

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
 Second Semester Examination 2023/2024 Session
 GET 321 Engineering Mathematics IV.
 Time Allotted: 3 Hours

Instruction: Answer four questions in all. Question 1 is compulsory

QUESTION 1.

(a) The following data gives the worldwide number of fatal airline accidents of commercially scheduled air transport in the years from 1997 to 2005.

Year	1997	1998	1999	2000	2001	2002	2003	2004	2005
Accidents	25	20	21	18	13	13	7	9	18

Calculate (i). the mean (ii). Sample variance (iii). Sample standard [15 marks]

(b) The following table gives data regarding leading sources of technology transfer (in percentage):

Countries	UK	Drmk	Ca	US	Sp	Jp	Ge	ODC	DC
Percentage	4	10	5	14	6	10	17	19	15

Represent the data by a suitable pie diagram. [10 marks]

QUESTION 2.

The oil seal of the Perkins generator serving the FUPRE, College of Engineering and Technology is known to have a life exponential distribution with a constant failure rate of 0.02×10^{-4} per hour. Using the above information (i) what is the probability that a given seal will last beyond 30,000 hours. (ii) what is the mean time to failure of the seal? (iii) what is the reliability at the mean time? (iv) if the reliability at its design life is given as 90%, what is the recommended design life? [15 marks]

QUESTION 3

(a) A class in probability theory consist of 6 men and 4 women. An exam is given, and the students are ranked according to their performance. Assume that no two students obtain the same score. (i). How many different ranks are possible? (ii). If all rankings are considered equally likely, what is the possibility that women received the top 4 scores?

(b) The probability that all lecturers in FUPRE own a car is 0.87, and the probability that the air conditioners fitted in their offices are in good order is 0.70. Also, the probability that a lecturer both owns a car and has a well-functioning air conditioner in their office is 0.65. What is the probability that a randomly selected lecturer owns a car or has a well-functioning office air conditioner?[15 marks]

QUESTION 4.

Draw a percentage curve for the following distribution of marks obtained by 700 students in an examination:

Marks	0-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89
No. of students	9	42	61	140	250	102	71	23	2

CF

Find from the graph (i) the marks at the 20th percentile, and (ii) the percentile equivalent to a mark of 65. [15 marks]

QUESTION 5.

The manufacturer of a new fiberglass claims that its average life will be at least 40,000 miles. To verify this, claim a sample of 12 liters is tested, with their lifetimes (in 1,000s of miles) being as follows:

Liters	1	2	3	4	5	6	7	8	9	10	11	12
Life	36.1	40.2	33.8	38.5	42	35.8	37	41	36.8	37.2	33	36

Test the manufacturer's claim at the 5 percent level of significance [15 marks]

mean
Standard

QUESTION 6.

The raw material used in the production of a synthetic fiber is stored in a location without a humidity control. Measurements of the relative humidity in the storage location and moisture content of a sample of the raw materials were taken over 15 days within the following data (in percentages) resulting.

Relative humidity	46	53	29	61	36	39	47	49	52	38	55	32	57	54	44
Moisture content	12	15	7	17	10	11	11	12	14	9	16	8	18	14	12

Use the data to plot the regression graph and estimate the regression equation from your graph. [15 marks]



Paper 5 of 6 — GET 321 MECHANICAL ENGINEERING 2024 past questions

Level 300 • Second Semester • Handout

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GET 321/2024/3 630


FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
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Relative humidity	46	53	29	61	36	39	47	49	52	38	55	32	57	54	44
Moisture content	12	15	7	17	10	11	11	12	14	9	16	8	18	14	12

Use the data to plot the regression graph and estimate the regression equation from your graph. [15 marks]



Paper 6 of 6 — GET321

Level 300 • Second Semester • Handout

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z .00 .01 .02 .03 .04 .05 .06 .07 .08 .09

-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207

-0.3 .38209 .37828 .37448 .37070 .36693 .36317 .35942 .35569 .35197 .34827
 -0.2 .42074 .41683 .41294 .40905 .40517 .40129 .39743 .39358 .38974 .38591
 -0.1 .46017 .45620 .45224 .44828 .44433 .44038 .43644 .43251 .42858 .42465
 -0.0 .50000 .49601 .49202 .48803 .48405 .48006 .47608 .47210 .46812 .46414

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z .00 .01 .02 .03 .04 .05 .06 .07 .08 .09

0.0 .50000 .50399 .50798 .51197 .51595 .51994 .52392 .52790 .53188 .53586
 0.1 .53983 .54380 .54776 .55172 .55567 .55962 .56356 .56749 .57142 .57535
 0.2 .57926 .58317 .58706 .59095 .59483 .59871 .60257 .60642 .61026 .61409
 0.3 .61791 .62172 .62552 .62930 .63307 .63683 .64058 .64431 .64803 .65173
 0.4 .65542 .65910 .66276 .66640 .67003 .67364 .67724 .68082 .68439 .68793
 0.5 .69146 .69497 .69847 .70194 .70540 .70884 .71226 .71566 .71904 .72240
 0.6 .72575 .72907 .73237 .73565 .73891 .74215 .74537 .74857 .75175 .75490
 0.7 .75804 .76115 .76424 .76730 .77035 .77337 .77637 .77935 .78230 .78524
 0.8 .78814 .79103 .79389 .79673 .79955 .80234 .80511 .80785 .81057 .81327
 0.9 .81594 .81859 .82121 .82381 .82639 .82894 .83147 .83398 .83646 .83891
 1.0 .84134 .84375 .84614 .84849 .85083 .85314 .85543 .85769 .85993 .86214
 1.1 .86433 .86650 .86864 .87076 .87286 .87493 .87698 .87900 .88100 .88298
 1.2 .88493 .88686 .88877 .89065 .89251 .89435 .89617 .89796 .89973 .90147
 1.3 .90320 .90490 .90658 .90824 .90988 .91149 .91309 .91466 .91621 .91774
 1.4 .91924 .92073 .92220 .92364 .92507 .92647 .92785 .92922 .93056 .93189
 1.5 .93319 .93448 .93574 .93699 .93822 .93943 .94062 .94179 .94295 .94408
 1.6 .94520 .94630 .94738 .94845 .94950 .95053 .95154 .95254 .95352 .95449
 1.7 .95543 .95637 .95728 .95818 .95907 .95994 .96080 .96164 .96246 .96327
 1.8 .96407 .96485 .96562 .96638 .96712 .96784 .96856 .96926 .96995 .97062
 1.9 .97128 .97193 .97257 .97320 .97381 .97441 .97500 .97558 .97615 .97670
 2.0 .97725 .97778 .97831 .97882 .97932 .97982 .98030 .98077 .98124 .98169
 2.1 .98214 .98257 .98300 .98341 .98382 .98422 .98461 .98500 .98537 .98574
 2.2 .98610 .98645 .98679 .98713 .98745 .98778 .98809 .98840 .98870 .98899
 2.3 .98928 .98956 .98983 .99010 .99036 .99061 .99086 .99111 .99134 .99158
 2.4 .99180 .99202 .99224 .99245 .99266 .99286 .99305 .99324 .99343 .99361
 2.5 .99379 .99396 .99413 .99430 .99446 .99461 .99477 .99492 .99506 .99520
 2.6 .99534 .99547 .99560 .99573 .99585 .99598 .99609 .99621 .99632 .99643
 2.7 .99653 .99664 .99674 .99683 .99693 .99702 .99711 .99720 .99728 .99736
 2.8 .99744 .99752 .99760 .99767 .99774 .99781 .99788 .99795 .99801 .99807
 2.9 .99813 .99819 .99825 .99831 .99836 .99841 .99846 .99851 .99856 .99861
 3.0 .99865 .99869 .99874 .99878 .99882 .99886 .99889 .99893 .99896 .99900
 3.1 .99903 .99906 .99910 .99913 .99916 .99918 .99921 .99924 .99926 .99929
 3.2 .99931 .99934 .99936 .99938 .99940 .99942 .99944 .99946 .99948 .99950
 3.3 .99952 .99953 .99955 .99957 .99958 .99960 .99961 .99962 .99964 .99965
 3.4 .99966 .99968 .99969 .99970 .99971 .99972 .99973 .99974 .99975 .99976

3.5 .99977 .99978 .99978 .99979 .99980 .99981 .99981 .99982 .99983 .99983
 3.6 .99984 .99985 .99985 .99986 .99986 .99987 .99987 .99988 .99988 .99989
 3.7 .99989 .99990 .99990 .99990 .99991 .99991 .99992 .99992 .99992 .99992
 3.8 .99993 .99993 .99993 .99994 .99994 .99994 .99994 .99995 .99995 .99995
 3.9 .99995 .99995 .99996 .99996 .99996 .99996 .99996 .99996 .99997 .99997

Table of critical values for the F distribution (for use with ANOVA):

How to use this table:

There are two tables here. The first one gives critical values of F at the $p = 0.05$ level of significance.

The second table gives critical values of F at the $p = 0.01$ level of significance.

1. Obtain your F-ratio. This has (x,y) degrees of freedom associated with it.
2. Go along x columns, and down y rows. The point of intersection is your critical F-ratio.
3. If your obtained value of F is equal to or larger than this critical F-value, then your result is significant at that level of probability.

An example: I obtain an F ratio of 3.96 with (2, 24) degrees of freedom.

I go along 2 columns and down 24 rows. The critical value of F is 3.40. My obtained F-ratio is larger than this, and so I conclude that my obtained F-ratio is likely to occur by chance with a $p < .05$.

Critical values of F for the 0.05 significance level:

1 2 3 4 5 6 7 8 9 10
 1 161.45 199.50 215.71 224.58 230.16 233.99 236.77 238.88 240.54 241.88
 2 18.51 19.00 19.16 19.25 19.30 19.33 19.35 19.37 19.39 19.40
 3 10.13 9.55 9.28 9.12 9.01 8.94 8.89 8.85 8.81 8.79
 4 7.71 6.94 6.59 6.39 6.26 6.16 6.09 6.04 6.00 5.96
 5 6.61 5.79 5.41 5.19 5.05 4.95 4.88 4.82 4.77 4.74
 6 5.99 5.14 4.76 4.53 4.39 4.28 4.21 4.15 4.10 4.06
 7 5.59 4.74 4.35 4.12 3.97 3.87 3.79 3.73 3.68 3.64
 8 5.32 4.46 4.07 3.84 3.69 3.58 3.50 3.44 3.39 3.35
 9 5.12 4.26 3.86 3.63 3.48 3.37 3.29 3.23 3.18 3.14
 10 4.97 4.10 3.71 3.48 3.33 3.22 3.14 3.07 3.02 2.98
 11 4.84 3.98 3.59 3.36 3.20 3.10 3.01 2.95 2.90 2.85
 12 4.75 3.89 3.49 3.26 3.11 3.00 2.91 2.85 2.80 2.75
 13 4.67 3.81 3.41 3.18 3.03 2.92 2.83 2.77 2.71 2.67
 14 4.60 3.74 3.34 3.11 2.96 2.85 2.76 2.70 2.65 2.60
 15 4.54 3.68 3.29 3.06 2.90 2.79 2.71 2.64 2.59 2.54
 16 4.49 3.63 3.24 3.01 2.85 2.74 2.66 2.59 2.54 2.49
 17 4.45 3.59 3.20 2.97 2.81 2.70 2.61 2.55 2.49 2.45
 18 4.41 3.56 3.16 2.93 2.77 2.66 2.58 2.51 2.46 2.41
 19 4.38 3.52 3.13 2.90 2.74 2.63 2.54 2.48 2.42 2.38
 20 4.35 3.49 3.10 2.87 2.71 2.60 2.51 2.45 2.39 2.35

21 4.33 3.47 3.07 2.84 2.69 2.57 2.49 2.42 2.37 2.32
22 4.30 3.44 3.05 2.82 2.66 2.55 2.46 2.40 2.34 2.30
23 4.28 3.42 3.03 2.80 2.64 2.53 2.44 2.38 2.32 2.28
24 4.26 3.40 3.01 2.78 2.62 2.51 2.42 2.36 2.30 2.26
25 4.24 3.39 2.99 2.76 2.60 2.49 2.41 2.34 2.28 2.24
26 4.23 3.37 2.98 2.74 2.59 2.47 2.39 2.32 2.27 2.22
27 4.21 3.35 2.96 2.73 2.57 2.46 2.37 2.31 2.25 2.20
28 4.20 3.34 2.95 2.71 2.56 2.45 2.36 2.29 2.24 2.19
29 4.18 3.33 2.93 2.70 2.55 2.43 2.35 2.28 2.22 2.18
30 4.17 3.32 2.92 2.69 2.53 2.42 2.33 2.27 2.21 2.17
31 4.16 3.31 2.91 2.68 2.52 2.41 2.32 2.26 2.20 2.15
32 4.15 3.30 2.90 2.67 2.51 2.40 2.31 2.24 2.19 2.14
33 4.14 3.29 2.89 2.66 2.50 2.39 2.30 2.24 2.18 2.13
34 4.13 3.28 2.88 2.65 2.49 2.38 2.29 2.23 2.17 2.12
35 4.12 3.27 2.87 2.64 2.49 2.37 2.29 2.22 2.16 2.11
36 4.11 3.26 2.87 2.63 2.48 2.36 2.28 2.21 2.15 2.11
37 4.11 3.25 2.86 2.63 2.47 2.36 2.27 2.20 2.15 2.10
38 4.10 3.25 2.85 2.62 2.46 2.35 2.26 2.19 2.14 2.09
39 4.09 3.24 2.85 2.61 2.46 2.34 2.26 2.19 2.13 2.08
40 4.09 3.23 2.84 2.61 2.45 2.34 2.25 2.18 2.12 2.08
41 4.08 3.23 2.83 2.60 2.44 2.33 2.24 2.17 2.12 2.07
42 4.07 3.22 2.83 2.59 2.44 2.32 2.24 2.17 2.11 2.07
43 4.07 3.21 2.82 2.59 2.43 2.32 2.23 2.16 2.11 2.06
44 4.06 3.21 2.82 2.58 2.43 2.31 2.23 2.16 2.10 2.05
45 4.06 3.20 2.81 2.58 2.42 2.31 2.22 2.15 2.10 2.05
46 4.05 3.20 2.81 2.57 2.42 2.30 2.22 2.15 2.09 2.04
47 4.05 3.20 2.80 2.57 2.41 2.30 2.21 2.14 2.09 2.04
48 4.04 3.19 2.80 2.57 2.41 2.30 2.21 2.14 2.08 2.04
49 4.04 3.19 2.79 2.56 2.40 2.29 2.20 2.13 2.08 2.03
50 4.03 3.18 2.79 2.56 2.40 2.29 2.20 2.13 2.07 2.03
51 4.03 3.18 2.79 2.55 2.40 2.28 2.20 2.13 2.07 2.02
52 4.03 3.18 2.78 2.55 2.39 2.28 2.19 2.12 2.07 2.02
53 4.02 3.17 2.78 2.55 2.39 2.28 2.19 2.12 2.06 2.02
54 4.02 3.17 2.78 2.54 2.39 2.27 2.19 2.12 2.06 2.01
55 4.02 3.17 2.77 2.54 2.38 2.27 2.18 2.11 2.06 2.01
56 4.01 3.16 2.77 2.54 2.38 2.27 2.18 2.11 2.05 2.01
57 4.01 3.16 2.77 2.53 2.38 2.26 2.18 2.11 2.05 2.00
58 4.01 3.16 2.76 2.53 2.37 2.26 2.17 2.10 2.05 2.00
59 4.00 3.15 2.76 2.53 2.37 2.26 2.17 2.10 2.04 2.00
60 4.00 3.15 2.76 2.53 2.37 2.25 2.17 2.10 2.04 1.99
61 4.00 3.15 2.76 2.52 2.37 2.25 2.16 2.09 2.04 1.99

62 4.00 3.15 2.75 2.52 2.36 2.25 2.16 2.09 2.04 1.99
 63 3.99 3.14 2.75 2.52 2.36 2.25 2.16 2.09 2.03 1.99
 64 3.99 3.14 2.75 2.52 2.36 2.24 2.16 2.09 2.03 1.98
 65 3.99 3.14 2.75 2.51 2.36 2.24 2.15 2.08 2.03 1.98
 66 3.99 3.14 2.74 2.51 2.35 2.24 2.15 2.08 2.03 1.98
 67 3.98 3.13 2.74 2.51 2.35 2.24 2.15 2.08 2.02 1.98
 68 3.98 3.13 2.74 2.51 2.35 2.24 2.15 2.08 2.02 1.97
 69 3.98 3.13 2.74 2.51 2.35 2.23 2.15 2.08 2.02 1.97
 70 3.98 3.13 2.74 2.50 2.35 2.23 2.14 2.07 2.02 1.97
 71 3.98 3.13 2.73 2.50 2.34 2.23 2.14 2.07 2.02 1.97
 72 3.97 3.12 2.73 2.50 2.34 2.23 2.14 2.07 2.01 1.97
 73 3.97 3.12 2.73 2.50 2.34 2.23 2.14 2.07 2.01 1.96
 74 3.97 3.12 2.73 2.50 2.34 2.22 2.14 2.07 2.01 1.96
 75 3.97 3.12 2.73 2.49 2.34 2.22 2.13 2.06 2.01 1.96
 76 3.97 3.12 2.73 2.49 2.34 2.22 2.13 2.06 2.01 1.96
 77 3.97 3.12 2.72 2.49 2.33 2.22 2.13 2.06 2.00 1.96
 78 3.96 3.11 2.72 2.49 2.33 2.22 2.13 2.06 2.00 1.95
 79 3.96 3.11 2.72 2.49 2.33 2.22 2.13 2.06 2.00 1.95
 80 3.96 3.11 2.72 2.49 2.33 2.21 2.13 2.06 2.00 1.95
 81 3.96 3.11 2.72 2.48 2.33 2.21 2.13 2.06 2.00 1.95
 82 3.96 3.11 2.72 2.48 2.33 2.21 2.12 2.05 2.00 1.95
 83 3.96 3.11 2.72 2.48 2.32 2.21 2.12 2.05 2.00 1.95
 84 3.96 3.11 2.71 2.48 2.32 2.21 2.12 2.05 1.99 1.95
 85 3.95 3.10 2.71 2.48 2.32 2.21 2.12 2.05 1.99 1.94
 86 3.95 3.10 2.71 2.48 2.32 2.21 2.12 2.05 1.99 1.94
 87 3.95 3.10 2.71 2.48 2.32 2.21 2.12 2.05 1.99 1.94
 88 3.95 3.10 2.71 2.48 2.32 2.20 2.12 2.05 1.99 1.94
 89 3.95 3.10 2.71 2.47 2.32 2.20 2.11 2.04 1.99 1.94
 90 3.95 3.10 2.71 2.47 2.32 2.20 2.11 2.04 1.99 1.94
 91 3.95 3.10 2.71 2.47 2.32 2.20 2.11 2.04 1.98 1.94
 92 3.95 3.10 2.70 2.47 2.31 2.20 2.11 2.04 1.98 1.94
 93 3.94 3.09 2.70 2.47 2.31 2.20 2.11 2.04 1.98 1.93
 94 3.94 3.09 2.70 2.47 2.31 2.20 2.11 2.04 1.98 1.93
 95 3.94 3.09 2.70 2.47 2.31 2.20 2.11 2.04 1.98 1.93
 96 3.94 3.09 2.70 2.47 2.31 2.20 2.11 2.04 1.98 1.93
 97 3.94 3.09 2.70 2.47 2.31 2.19 2.11 2.04 1.98 1.93
 98 3.94 3.09 2.70 2.47 2.31 2.19 2.10 2.03 1.98 1.93
 99 3.94 3.09 2.70 2.46 2.31 2.19 2.10 2.03 1.98 1.93
 100 3.94 3.09 2.70 2.46 2.31 2.19 2.10 2.03 1.98 1.93

Critical values of F for the 0.01 significance level:

1 2 3 4 5 6 7 8 9 10

1 4052.19 4999.52 5403.34 5624.62 5763.65 5858.97 5928.33 5981.10 6022.50 6055.85
2 98.50 99.00 99.17 99.25 99.30 99.33 99.36 99.37 99.39 99.40
3 34.12 30.82 29.46 28.71 28.24 27.91 27.67 27.49 27.35 27.23
4 21.20 18.00 16.69 15.98 15.52 15.21 14.98 14.80 14.66 14.55
5 16.26 13.27 12.06 11.39 10.97 10.67 10.46 10.29 10.16 10.05
6 13.75 10.93 9.78 9.15 8.75 8.47 8.26 8.10 7.98 7.87
7 12.25 9.55 8.45 7.85 7.46 7.19 6.99 6.84 6.72 6.62
8 11.26 8.65 7.59 7.01 6.63 6.37 6.18 6.03 5.91 5.81
9 10.56 8.02 6.99 6.42 6.06 5.80 5.61 5.47 5.35 5.26
10 10.04 7.56 6.55 5.99 5.64 5.39 5.20 5.06 4.94 4.85
11 9.65 7.21 6.22 5.67 5.32 5.07 4.89 4.74 4.63 4.54
12 9.33 6.93 5.95 5.41 5.06 4.82 4.64 4.50 4.39 4.30
13 9.07 6.70 5.74 5.21 4.86 4.62 4.44 4.30 4.19 4.10
14 8.86 6.52 5.56 5.04 4.70 4.46 4.28 4.14 4.03 3.94
15 8.68 6.36 5.42 4.89 4.56 4.32 4.14 4.00 3.90 3.81
16 8.53 6.23 5.29 4.77 4.44 4.20 4.03 3.89 3.78 3.69
17 8.40 6.11 5.19 4.67 4.34 4.10 3.93 3.79 3.68 3.59
18 8.29 6.01 5.09 4.58 4.25 4.02 3.84 3.71 3.60 3.51
19 8.19 5.93 5.01 4.50 4.17 3.94 3.77 3.63 3.52 3.43
20 8.10 5.85 4.94 4.43 4.10 3.87 3.70 3.56 3.46 3.37
21 8.02 5.78 4.87 4.37 4.04 3.81 3.64 3.51 3.40 3.31
22 7.95 5.72 4.82 4.31 3.99 3.76 3.59 3.45 3.35 3.26
23 7.88 5.66 4.77 4.26 3.94 3.71 3.54 3.41 3.30 3.21
24 7.82 5.61 4.72 4.22 3.90 3.67 3.50 3.36 3.26 3.17
25 7.77 5.57 4.68 4.18 3.86 3.63 3.46 3.32 3.22 3.13
26 7.72 5.53 4.64 4.14 3.82 3.59 3.42 3.29 3.18 3.09
27 7.68 5.49 4.60 4.11 3.79 3.56 3.39 3.26 3.15 3.06
28 7.64 5.45 4.57 4.07 3.75 3.53 3.36 3.23 3.12 3.03
29 7.60 5.42 4.54 4.05 3.73 3.50 3.33 3.20 3.09 3.01
30 7.56 5.39 4.51 4.02 3.70 3.47 3.31 3.17 3.07 2.98
31 7.53 5.36 4.48 3.99 3.68 3.45 3.28 3.15 3.04 2.96
32 7.50 5.34 4.46 3.97 3.65 3.43 3.26 3.13 3.02 2.93
33 7.47 5.31 4.44 3.95 3.63 3.41 3.24 3.11 3.00 2.91
34 7.44 5.29 4.42 3.93 3.61 3.39 3.22 3.09 2.98 2.89
35 7.42 5.27 4.40 3.91 3.59 3.37 3.20 3.07 2.96 2.88
36 7.40 5.25 4.38 3.89 3.57 3.35 3.18 3.05 2.95 2.86
37 7.37 5.23 4.36 3.87 3.56 3.33 3.17 3.04 2.93 2.84
38 7.35 5.21 4.34 3.86 3.54 3.32 3.15 3.02 2.92 2.83
39 7.33 5.19 4.33 3.84 3.53 3.31 3.14 3.01 2.90 2.81
40 7.31 5.18 4.31 3.83 3.51 3.29 3.12 2.99 2.89 2.80
41 7.30 5.16 4.30 3.82 3.50 3.28 3.11 2.98 2.88 2.79

42 7.28 5.15 4.29 3.80 3.49 3.27 3.10 2.97 2.86 2.78
43 7.26 5.14 4.27 3.79 3.48 3.25 3.09 2.96 2.85 2.76
44 7.25 5.12 4.26 3.78 3.47 3.24 3.08 2.95 2.84 2.75
45 7.23 5.11 4.25 3.77 3.45 3.23 3.07 2.94 2.83 2.74
46 7.22 5.10 4.24 3.76 3.44 3.22 3.06 2.93 2.82 2.73
47 7.21 5.09 4.23 3.75 3.43 3.21 3.05 2.92 2.81 2.72
48 7.19 5.08 4.22 3.74 3.43 3.20 3.04 2.91 2.80 2.72
49 7.18 5.07 4.21 3.73 3.42 3.20 3.03 2.90 2.79 2.71
50 7.17 5.06 4.20 3.72 3.41 3.19 3.02 2.89 2.79 2.70
51 7.16 5.05 4.19 3.71 3.40 3.18 3.01 2.88 2.78 2.69
52 7.15 5.04 4.18 3.70 3.39 3.17 3.01 2.87 2.77 2.68
53 7.14 5.03 4.17 3.70 3.38 3.16 3.00 2.87 2.76 2.68
54 7.13 5.02 4.17 3.69 3.38 3.16 2.99 2.86 2.76 2.67
55 7.12 5.01 4.16 3.68 3.37 3.15 2.98 2.85 2.75 2.66
56 7.11 5.01 4.15 3.67 3.36 3.14 2.98 2.85 2.74 2.66
57 7.10 5.00 4.15 3.67 3.36 3.14 2.97 2.84 2.74 2.65
58 7.09 4.99 4.14 3.66 3.35 3.13 2.97 2.84 2.73 2.64
59 7.09 4.98 4.13 3.66 3.35 3.12 2.96 2.83 2.72 2.64
60 7.08 4.98 4.13 3.65 3.34 3.12 2.95 2.82 2.72 2.63
61 7.07 4.97 4.12 3.64 3.33 3.11 2.95 2.82 2.71 2.63
62 7.06 4.97 4.11 3.64 3.33 3.11 2.94 2.81 2.71 2.62
63 7.06 4.96 4.11 3.63 3.32 3.10 2.94 2.81 2.70 2.62
64 7.05 4.95 4.10 3.63 3.32 3.10 2.93 2.80 2.70 2.61
65 7.04 4.95 4.10 3.62 3.31 3.09 2.93 2.80 2.69 2.61
66 7.04 4.94 4.09 3.62 3.31 3.09 2.92 2.79 2.69 2.60
67 7.03 4.94 4.09 3.61 3.30 3.08 2.92 2.79 2.68 2.60
68 7.02 4.93 4.08 3.61 3.30 3.08 2.91 2.79 2.68 2.59
69 7.02 4.93 4.08 3.60 3.30 3.08 2.91 2.78 2.68 2.59
70 7.01 4.92 4.07 3.60 3.29 3.07 2.91 2.78 2.67 2.59
71 7.01 4.92 4.07 3.60 3.29 3.07 2.90 2.77 2.67 2.58
72 7.00 4.91 4.07 3.59 3.28 3.06 2.90 2.77 2.66 2.58
73 7.00 4.91 4.06 3.59 3.28 3.06 2.90 2.77 2.66 2.57
74 6.99 4.90 4.06 3.58 3.28 3.06 2.89 2.76 2.66 2.57
75 6.99 4.90 4.05 3.58 3.27 3.05 2.89 2.76 2.65 2.57
76 6.98 4.90 4.05 3.58 3.27 3.05 2.88 2.76 2.65 2.56
77 6.98 4.89 4.05 3.57 3.27 3.05 2.88 2.75 2.65 2.56
78 6.97 4.89 4.04 3.57 3.26 3.04 2.88 2.75 2.64 2.56
79 6.97 4.88 4.04 3.57 3.26 3.04 2.87 2.75 2.64 2.55
80 6.96 4.88 4.04 3.56 3.26 3.04 2.87 2.74 2.64 2.55
81 6.96 4.88 4.03 3.56 3.25 3.03 2.87 2.74 2.63 2.55
82 6.95 4.87 4.03 3.56 3.25 3.03 2.87 2.74 2.63 2.55

83	6.95	4.87	4.03	3.55	3.25	3.03	2.86	2.73	2.63	2.54
84	6.95	4.87	4.02	3.55	3.24	3.03	2.86	2.73	2.63	2.54
85	6.94	4.86	4.02	3.55	3.24	3.02	2.86	2.73	2.62	2.54
86	6.94	4.86	4.02	3.55	3.24	3.02	2.85	2.73	2.62	2.53
87	6.94	4.86	4.02	3.54	3.24	3.02	2.85	2.72	2.62	2.53
88	6.93	4.86	4.01	3.54	3.23	3.01	2.85	2.72	2.62	2.53
89	6.93	4.85	4.01	3.54	3.23	3.01	2.85	2.72	2.61	2.53
90	6.93	4.85	4.01	3.54	3.23	3.01	2.85	2.72	2.61	2.52
91	6.92	4.85	4.00	3.53	3.23	3.01	2.84	2.71	2.61	2.52
92	6.92	4.84	4.00	3.53	3.22	3.00	2.84	2.71	2.61	2.52
93	6.92	4.84	4.00	3.53	3.22	3.00	2.84	2.71	2.60	2.52
94	6.91	4.84	4.00	3.53	3.22	3.00	2.84	2.71	2.60	2.52
95	6.91	4.84	4.00	3.52	3.22	3.00	2.83	2.70	2.60	2.51
96	6.91	4.83	3.99	3.52	3.21	3.00	2.83	2.70	2.60	2.51
97	6.90	4.83	3.99	3.52	3.21	2.99	2.83	2.70	2.60	2.51
98	6.90	4.83	3.99	3.52	3.21	2.99	2.83	2.70	2.59	2.51
99	6.90	4.83	3.99	3.52	3.21	2.99	2.83	2.70	2.59	2.51
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.50

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T-12 Tables

Probability p

Table entry for p is the
critical value F^* with
probability p lying to
its right. F^*

TABLE E

F critical values

Degrees of freedom in the numerator

p 1 2 3 4 5 6 7 8 9

.100	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86
.050	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54
1 .025	647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28
.010	4052.2	4999.5	5403.4	5624.6	5763.6	5859.0	5928.4	5981.1	6022.5
.001	405284	500000	540379	562500	576405	585937	592873	598144	602284
.100	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38
.050	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
2 .025	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39

.010	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39
.001	998.50	999.00	999.17	999.25	999.30	999.33	999.36	999.37	999.39
.100	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24
Degrees of freedom in the denominator									
.050	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
3 .025	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47
.010	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35
.001	167.03	148.50	141.11	137.10	134.58	132.85	131.58	130.62	129.86
.100	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94
.050	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
4 .025	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90
.010	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66
.001	74.14	61.25	56.18	53.44	51.71	50.53	49.66	49.00	48.47
.100	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32
.050	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
5 .025	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68
.010	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16
.001	47.18	37.12	33.20	31.09	29.75	28.83	28.16	27.65	27.24
.100	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96
.050	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
6 .025	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52
.010	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98
.001	35.51	27.00	23.70	21.92	20.80	20.03	19.46	19.03	18.69
.100	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72
.050	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
7 .025	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82
.010	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72
.001	29.25	21.69	18.77	17.20	16.21	15.52	15.02	14.63	14.33

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Tables T-13

Probability p

Table entry for p is the
critical value F^* with
probability p lying to
its right. F^*

TABLE E

F critical values (continued)

Degrees of freedom in the numerator

10 12 15 20 25 30 40 50 60 120 1000
 60.19 60.71 61.22 61.74 62.05 62.26 62.53 62.69 62.79 63.06 63.30
 241.88 243.91 245.95 248.01 249.26 250.10 251.14 251.77 252.20 253.25 254.19
 968.63 976.71 984.87 993.10 998.08 1001.4 1005.6 1008.1 1009.8 1014.0 1017.7
 6055.8 6106.3 6157.3 6208.7 6239.8 6260.6 6286.8 6302.5 6313.0 6339.4 6362.7
 605621 610668 615764 620908 624017 626099 628712 630285 631337 633972 636301
 9.39 9.41 9.42 9.44 9.45 9.46 9.47 9.47 9.47 9.48 9.49
 19.40 19.41 19.43 19.45 19.46 19.46 19.47 19.48 19.48 19.49 19.49
 39.40 39.41 39.43 39.45 39.46 39.46 39.47 39.48 39.48 39.49 39.50
 99.40 99.42 99.43 99.45 99.46 99.47 99.47 99.48 99.48 99.49 99.50
 999.40 999.42 999.43 999.45 999.46 999.47 999.47 999.48 999.48 999.49 999.50
 5.23 5.22 5.20 5.18 5.17 5.17 5.16 5.15 5.15 5.14 5.13
 8.79 8.74 8.70 8.66 8.63 8.62 8.59 8.58 8.57 8.55 8.53
 14.42 14.34 14.25 14.17 14.12 14.08 14.04 14.01 13.99 13.95 13.91
 27.23 27.05 26.87 26.69 26.58 26.50 26.41 26.35 26.32 26.22 26.14
 129.25 128.32 127.37 126.42 125.84 125.45 124.96 124.66 124.47 123.97 123.53
 3.92 3.90 3.87 3.84 3.83 3.82 3.80 3.80 3.79 3.78 3.76
 5.96 5.91 5.86 5.80 5.77 5.75 5.72 5.70 5.69 5.66 5.63
 8.84 8.75 8.66 8.56 8.50 8.46 8.41 8.38 8.36 8.31 8.26
 14.55 14.37 14.20 14.02 13.91 13.84 13.75 13.69 13.65 13.56 13.47
 48.05 47.41 46.76 46.10 45.70 45.43 45.09 44.88 44.75 44.40 44.09
 3.30 3.27 3.24 3.21 3.19 3.17 3.16 3.15 3.14 3.12 3.11
 4.74 4.68 4.62 4.56 4.52 4.50 4.46 4.44 4.43 4.40 4.37
 6.62 6.52 6.43 6.33 6.27 6.23 6.18 6.14 6.12 6.07 6.02
 10.05 9.89 9.72 9.55 9.45 9.38 9.29 9.24 9.20 9.11 9.03
 26.92 26.42 25.91 25.39 25.08 24.87 24.60 24.44 24.33 24.06 23.82
 2.94 2.90 2.87 2.84 2.81 2.80 2.78 2.77 2.76 2.74 2.72
 4.06 4.00 3.94 3.87 3.83 3.81 3.77 3.75 3.74 3.70 3.67
 5.46 5.37 5.27 5.17 5.11 5.07 5.01 4.98 4.96 4.90 4.86
 7.87 7.72 7.56 7.40 7.30 7.23 7.14 7.09 7.06 6.97 6.89
 18.41 17.99 17.56 17.12 16.85 16.67 16.44 16.31 16.21 15.98 15.77
 2.70 2.67 2.63 2.59 2.57 2.56 2.54 2.52 2.51 2.49 2.47
 3.64 3.57 3.51 3.44 3.40 3.38 3.34 3.32 3.30 3.27 3.23
 4.76 4.67 4.57 4.47 4.40 4.36 4.31 4.28 4.25 4.20 4.15
 6.62 6.47 6.31 6.16 6.06 5.99 5.91 5.86 5.82 5.74 5.66
 14.08 13.71 13.32 12.93 12.69 12.53 12.33 12.20 12.12 11.91 11.72

(Continued)

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TABLE E

F critical values (continued)

Degrees of freedom in the numerator

p 1 2 3 4 5 6 7 8 9

.100 3.46 3.11 2.92 2.81 2.73 2.67 2.62 2.59 2.56
 .050 5.32 4.46 4.07 3.84 3.69 3.58 3.50 3.44 3.39
 8 .025 7.57 6.06 5.42 5.05 4.82 4.65 4.53 4.43 4.36
 .010 11.26 8.65 7.59 7.01 6.63 6.37 6.18 6.03 5.91
 .001 25.41 18.49 15.83 14.39 13.48 12.86 12.40 12.05 11.77

.100 3.36 3.01 2.81 2.69 2.61 2.55 2.51 2.47 2.44
 .050 5.12 4.26 3.86 3.63 3.48 3.37 3.29 3.23 3.18
 9 .025 7.21 5.71 5.08 4.72 4.48 4.32 4.20 4.10 4.03
 .010 10.56 8.02 6.99 6.42 6.06 5.80 5.61 5.47 5.35
 .001 22.86 16.39 13.90 12.56 11.71 11.13 10.70 10.37 10.11

.100 3.29 2.92 2.73 2.61 2.52 2.46 2.41 2.38 2.35
 .050 4.96 4.10 3.71 3.48 3.33 3.22 3.14 3.07 3.02
 10 .025 6.94 5.46 4.83 4.47 4.24 4.07 3.95 3.85 3.78
 .010 10.04 7.56 6.55 5.99 5.64 5.39 5.20 5.06 4.94
 .001 21.04 14.91 12.55 11.28 10.48 9.93 9.52 9.20 8.96

.100 3.23 2.86 2.66 2.54 2.45 2.39 2.34 2.30 2.27
 .050 4.84 3.98 3.59 3.36 3.20 3.09 3.01 2.95 2.90
 11 .025 6.72 5.26 4.63 4.28 4.04 3.88 3.76 3.66 3.59

Degrees of freedom in the denominator

.010 9.65 7.21 6.22 5.67 5.32 5.07 4.89 4.74 4.63
 .001 19.69 13.81 11.56 10.35 9.58 9.05 8.66 8.35 8.12

.100 3.18 2.81 2.61 2.48 2.39 2.33 2.28 2.24 2.21
 .050 4.75 3.89 3.49 3.26 3.11 3.00 2.91 2.85 2.80
 12 .025 6.55 5.10 4.47 4.12 3.89 3.73 3.61 3.51 3.44
 .010 9.33 6.93 5.95 5.41 5.06 4.82 4.64 4.50 4.39
 .001 18.64 12.97 10.80 9.63 8.89 8.38 8.00 7.71 7.48

.100 3.14 2.76 2.56 2.43 2.35 2.28 2.23 2.20 2.16
 .050 4.67 3.81 3.41 3.18 3.03 2.92 2.83 2.77 2.71
 13 .025 6.41 4.97 4.35 4.00 3.77 3.60 3.48 3.39 3.31
 .010 9.07 6.70 5.74 5.21 4.86 4.62 4.44 4.30 4.19
 .001 17.82 12.31 10.21 9.07 8.35 7.86 7.49 7.21 6.98

.100 3.10 2.73 2.52 2.39 2.31 2.24 2.19 2.15 2.12
 .050 4.60 3.74 3.34 3.11 2.96 2.85 2.76 2.70 2.65
 14 .025 6.30 4.86 4.24 3.89 3.66 3.50 3.38 3.29 3.21
 .010 8.86 6.51 5.56 5.04 4.69 4.46 4.28 4.14 4.03

.001 17.14 11.78 9.73 8.62 7.92 7.44 7.08 6.80 6.58
 .100 3.07 2.70 2.49 2.36 2.27 2.21 2.16 2.12 2.09
 .050 4.54 3.68 3.29 3.06 2.90 2.79 2.71 2.64 2.59
 15 .025 6.20 4.77 4.15 3.80 3.58 3.41 3.29 3.20 3.12
 .010 8.68 6.36 5.42 4.89 4.56 4.32 4.14 4.00 3.89
 .001 16.59 11.34 9.34 8.25 7.57 7.09 6.74 6.47 6.26
 .100 3.05 2.67 2.46 2.33 2.24 2.18 2.13 2.09 2.06
 .050 4.49 3.63 3.24 3.01 2.85 2.74 2.66 2.59 2.54
 16 .025 6.12 4.69 4.08 3.73 3.50 3.34 3.22 3.12 3.05
 .010 8.53 6.23 5.29 4.77 4.44 4.20 4.03 3.89 3.78
 .001 16.12 10.97 9.01 7.94 7.27 6.80 6.46 6.19 5.98
 .100 3.03 2.64 2.44 2.31 2.22 2.15 2.10 2.06 2.03
 .050 4.45 3.59 3.20 2.96 2.81 2.70 2.61 2.55 2.49
 17 .025 6.04 4.62 4.01 3.66 3.44 3.28 3.16 3.06 2.98
 .010 8.40 6.11 5.19 4.67 4.34 4.10 3.93 3.79 3.68
 .001 15.72 10.66 8.73 7.68 7.02 6.56 6.22 5.96 5.75

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Tables T-15

TABLE E

F critical values (continued)

Degrees of freedom in the numerator

10 12 15 20 25 30 40 50 60 120 1000
 2.54 2.50 2.46 2.42 2.40 2.38 2.36 2.35 2.34 2.32 2.30
 3.35 3.28 3.22 3.15 3.11 3.08 3.04 3.02 3.01 2.97 2.93
 4.30 4.20 4.10 4.00 3.94 3.89 3.84 3.81 3.78 3.73 3.68
 5.81 5.67 5.52 5.36 5.26 5.20 5.12 5.07 5.03 4.95 4.87
 11.54 11.19 10.84 10.48 10.26 10.11 9.92 9.80 9.73 9.53 9.36
 2.42 2.38 2.34 2.30 2.27 2.25 2.23 2.22 2.21 2.18 2.16
 3.14 3.07 3.01 2.94 2.89 2.86 2.83 2.80 2.79 2.75 2.71
 3.96 3.87 3.77 3.67 3.60 3.56 3.51 3.47 3.45 3.39 3.34
 5.26 5.11 4.96 4.81 4.71 4.65 4.57 4.52 4.48 4.40 4.32
 9.89 9.57 9.24 8.90 8.69 8.55 8.37 8.26 8.19 8.00 7.84
 2.32 2.28 2.24 2.20 2.17 2.16 2.13 2.12 2.11 2.08 2.06
 2.98 2.91 2.85 2.77 2.73 2.70 2.66 2.64 2.62 2.58 2.54
 3.72 3.62 3.52 3.42 3.35 3.31 3.26 3.22 3.20 3.14 3.09
 4.85 4.71 4.56 4.41 4.31 4.25 4.17 4.12 4.08 4.00 3.92
 8.75 8.45 8.13 7.80 7.60 7.47 7.30 7.19 7.12 6.94 6.78
 2.25 2.21 2.17 2.12 2.10 2.08 2.05 2.04 2.03 2.00 1.98

2.85 2.79 2.72 2.65 2.60 2.57 2.53 2.51 2.49 2.45 2.41
 3.53 3.43 3.33 3.23 3.16 3.12 3.06 3.03 3.00 2.94 2.89
 4.54 4.40 4.25 4.10 4.01 3.94 3.86 3.81 3.78 3.69 3.61
 7.92 7.63 7.32 7.01 6.81 6.68 6.52 6.42 6.35 6.18 6.02
 2.19 2.15 2.10 2.06 2.03 2.01 1.99 1.97 1.96 1.93 1.91
 2.75 2.69 2.62 2.54 2.50 2.47 2.43 2.40 2.38 2.34 2.30
 3.37 3.28 3.18 3.07 3.01 2.96 2.91 2.87 2.85 2.79 2.73
 4.30 4.16 4.01 3.86 3.76 3.70 3.62 3.57 3.54 3.45 3.37
 7.29 7.00 6.71 6.40 6.22 6.09 5.93 5.83 5.76 5.59 5.44
 2.14 2.10 2.05 2.01 1.98 1.96 1.93 1.92 1.90 1.88 1.85
 2.67 2.60 2.53 2.46 2.41 2.38 2.34 2.31 2.30 2.25 2.21
 3.25 3.15 3.05 2.95 2.88 2.84 2.78 2.74 2.72 2.66 2.60
 4.10 3.96 3.82 3.66 3.57 3.51 3.43 3.38 3.34 3.25 3.18
 6.80 6.52 6.23 5.93 5.75 5.63 5.47 5.37 5.30 5.14 4.99
 2.10 2.05 2.01 1.96 1.93 1.91 1.89 1.87 1.86 1.83 1.80
 2.60 2.53 2.46 2.39 2.34 2.31 2.27 2.24 2.22 2.18 2.14
 3.15 3.05 2.95 2.84 2.78 2.73 2.67 2.64 2.61 2.55 2.50
 3.94 3.80 3.66 3.51 3.41 3.35 3.27 3.22 3.18 3.09 3.02
 6.40 6.13 5.85 5.56 5.38 5.25 5.10 5.00 4.94 4.77 4.62
 2.06 2.02 1.97 1.92 1.89 1.87 1.85 1.83 1.82 1.79 1.76
 2.54 2.48 2.40 2.33 2.28 2.25 2.20 2.18 2.16 2.11 2.07
 3.06 2.96 2.86 2.76 2.69 2.64 2.59 2.55 2.52 2.46 2.40
 3.80 3.67 3.52 3.37 3.28 3.21 3.13 3.08 3.05 2.96 2.88
 6.08 5.81 5.54 5.25 5.07 4.95 4.80 4.70 4.64 4.47 4.33
 2.03 1.99 1.94 1.89 1.86 1.84 1.81 1.79 1.78 1.75 1.72
 2.49 2.42 2.35 2.28 2.23 2.19 2.15 2.12 2.11 2.06 2.02
 2.99 2.89 2.79 2.68 2.61 2.57 2.51 2.47 2.45 2.38 2.32
 3.69 3.55 3.41 3.26 3.16 3.10 3.02 2.97 2.93 2.84 2.76
 5.81 5.55 5.27 4.99 4.82 4.70 4.54 4.45 4.39 4.23 4.08
 2.00 1.96 1.91 1.86 1.83 1.81 1.78 1.76 1.75 1.72 1.69
 2.45 2.38 2.31 2.23 2.18 2.15 2.10 2.08 2.06 2.01 1.97
 2.92 2.82 2.72 2.62 2.55 2.50 2.44 2.41 2.38 2.32 2.26
 3.59 3.46 3.31 3.16 3.07 3.00 2.92 2.87 2.83 2.75 2.66
 5.58 5.32 5.05 4.78 4.60 4.48 4.33 4.24 4.18 4.02 3.87

(Continued)

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TABLE E

F critical values (continued)

Degrees of freedom in the numerator

p 1 2 3 4 5 6 7 8 9

.100	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04	2.00
.050	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
18 .025	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93
.010	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60
.001	15.38	10.39	8.49	7.46	6.81	6.35	6.02	5.76	5.56
.100	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02	1.98
.050	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
19 .025	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88
.010	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52
.001	15.08	10.16	8.28	7.27	6.62	6.18	5.85	5.59	5.39
.100	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00	1.96
.050	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
20 .025	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84
.010	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46
.001	14.82	9.95	8.10	7.10	6.46	6.02	5.69	5.44	5.24
.100	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98	1.95
.050	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37
21 .025	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80

Degrees of freedom in the denominator

.010	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40
.001	14.59	9.77	7.94	6.95	6.32	5.88	5.56	5.31	5.11
.100	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97	1.93
.050	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
22 .025	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76
.010	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35
.001	14.38	9.61	7.80	6.81	6.19	5.76	5.44	5.19	4.99
.100	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95	1.92
.050	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32
23 .025	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73
.010	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30
.001	14.20	9.47	7.67	6.70	6.08	5.65	5.33	5.09	4.89
.100	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94	1.91
.050	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
24 .025	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70
.010	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26
.001	14.03	9.34	7.55	6.59	5.98	5.55	5.23	4.99	4.80
.100	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93	1.89

.050 4.24 3.39 2.99 2.76 2.60 2.49 2.40 2.34 2.28
 25 .025 5.69 4.29 3.69 3.35 3.13 2.97 2.85 2.75 2.68
 .010 7.77 5.57 4.68 4.18 3.85 3.63 3.46 3.32 3.22
 .001 13.88 9.22 7.45 6.49 5.89 5.46 5.15 4.91 4.71
 .100 2.91 2.52 2.31 2.17 2.08 2.01 1.96 1.92 1.88
 .050 4.23 3.37 2.98 2.74 2.59 2.47 2.39 2.32 2.27
 26 .025 5.66 4.27 3.67 3.33 3.10 2.94 2.82 2.73 2.65
 .010 7.72 5.53 4.64 4.14 3.82 3.59 3.42 3.29 3.18
 .001 13.74 9.12 7.36 6.41 5.80 5.38 5.07 4.83 4.64
 .100 2.90 2.51 2.30 2.17 2.07 2.00 1.95 1.91 1.87
 .050 4.21 3.35 2.96 2.73 2.57 2.46 2.37 2.31 2.25
 27 .025 5.63 4.24 3.65 3.31 3.08 2.92 2.80 2.71 2.63
 .010 7.68 5.49 4.60 4.11 3.78 3.56 3.39 3.26 3.15
 .001 13.61 9.02 7.27 6.33 5.73 5.31 5.00 4.76 4.57

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Tables T-17

TABLE E

F critical values (continued)

Degrees of freedom in the numerator

10 12 15 20 25 30 40 50 60 120 1000

1.98 1.93 1.89 1.84 1.80 1.78 1.75 1.74 1.72 1.69 1.66
 2.41 2.34 2.27 2.19 2.14 2.11 2.06 2.04 2.02 1.97 1.92
 2.87 2.77 2.67 2.56 2.49 2.44 2.38 2.35 2.32 2.26 2.20
 3.51 3.37 3.23 3.08 2.98 2.92 2.84 2.78 2.75 2.66 2.58
 5.39 5.13 4.87 4.59 4.42 4.30 4.15 4.06 4.00 3.84 3.69
 1.96 1.91 1.86 1.81 1.78 1.76 1.73 1.71 1.70 1.67 1.64
 2.38 2.31 2.23 2.16 2.11 2.07 2.03 2.00 1.98 1.93 1.88
 2.82 2.72 2.62 2.51 2.44 2.39 2.33 2.30 2.27 2.20 2.14
 3.43 3.30 3.15 3.00 2.91 2.84 2.76 2.71 2.67 2.58 2.50
 5.22 4.97 4.70 4.43 4.26 4.14 3.99 3.90 3.84 3.68 3.53
 1.94 1.89 1.84 1.79 1.76 1.74 1.71 1.69 1.68 1.64 1.61
 2.35 2.28 2.20 2.12 2.07 2.04 1.99 1.97 1.95 1.90 1.85
 2.77 2.68 2.57 2.46 2.40 2.35 2.29 2.25 2.22 2.16 2.09
 3.37 3.23 3.09 2.94 2.84 2.78 2.69 2.64 2.61 2.52 2.43
 5.08 4.82 4.56 4.29 4.12 4.00 3.86 3.77 3.70 3.54 3.40
 1.92 1.87 1.83 1.78 1.74 1.72 1.69 1.67 1.66 1.62 1.59
 2.32 2.25 2.18 2.10 2.05 2.01 1.96 1.94 1.92 1.87 1.82
 2.73 2.64 2.53 2.42 2.36 2.31 2.25 2.21 2.18 2.11 2.05
 3.31 3.17 3.03 2.88 2.79 2.72 2.64 2.58 2.55 2.46 2.37

4.95 4.70 4.44 4.17 4.00 3.88 3.74 3.64 3.58 3.42 3.28
 1.90 1.86 1.81 1.76 1.73 1.70 1.67 1.65 1.64 1.60 1.57
 2.30 2.23 2.15 2.07 2.02 1.98 1.94 1.91 1.89 1.84 1.79
 2.70 2.60 2.50 2.39 2.32 2.27 2.21 2.17 2.14 2.08 2.01
 3.26 3.12 2.98 2.83 2.73 2.67 2.58 2.53 2.50 2.40 2.32
 4.83 4.58 4.33 4.06 3.89 3.78 3.63 3.54 3.48 3.32 3.17
 1.89 1.84 1.80 1.74 1.71 1.69 1.66 1.64 1.62 1.59 1.55
 2.27 2.20 2.13 2.05 2.00 1.96 1.91 1.88 1.86 1.81 1.76
 2.67 2.57 2.47 2.36 2.29 2.24 2.18 2.14 2.11 2.04 1.98
 3.21 3.07 2.93 2.78 2.69 2.62 2.54 2.48 2.45 2.35 2.27
 4.73 4.48 4.23 3.96 3.79 3.68 3.53 3.44 3.38 3.22 3.08
 1.88 1.83 1.78 1.73 1.70 1.67 1.64 1.62 1.61 1.57 1.54
 2.25 2.18 2.11 2.03 1.97 1.94 1.89 1.86 1.84 1.79 1.74
 2.64 2.54 2.44 2.33 2.26 2.21 2.15 2.11 2.08 2.01 1.94
 3.17 3.03 2.89 2.74 2.64 2.58 2.49 2.44 2.40 2.31 2.22
 4.64 4.39 4.14 3.87 3.71 3.59 3.45 3.36 3.29 3.14 2.99
 1.87 1.82 1.77 1.72 1.68 1.66 1.63 1.61 1.59 1.56 1.52
 2.24 2.16 2.09 2.01 1.96 1.92 1.87 1.84 1.82 1.77 1.72
 2.61 2.51 2.41 2.30 2.23 2.18 2.12 2.08 2.05 1.98 1.91
 3.13 2.99 2.85 2.70 2.60 2.54 2.45 2.40 2.36 2.27 2.18
 4.56 4.31 4.06 3.79 3.63 3.52 3.37 3.28 3.22 3.06 2.91
 1.86 1.81 1.76 1.71 1.67 1.65 1.61 1.59 1.58 1.54 1.51
 2.22 2.15 2.07 1.99 1.94 1.90 1.85 1.82 1.80 1.75 1.70
 2.59 2.49 2.39 2.28 2.21 2.16 2.09 2.05 2.03 1.95 1.89
 3.09 2.96 2.81 2.66 2.57 2.50 2.42 2.36 2.33 2.23 2.14
 4.48 4.24 3.99 3.72 3.56 3.44 3.30 3.21 3.15 2.99 2.84
 1.85 1.80 1.75 1.70 1.66 1.64 1.60 1.58 1.57 1.53 1.50
 2.20 2.13 2.06 1.97 1.92 1.88 1.84 1.81 1.79 1.73 1.68
 2.57 2.47 2.36 2.25 2.18 2.13 2.07 2.03 2.00 1.93 1.86
 3.06 2.93 2.78 2.63 2.54 2.47 2.38 2.33 2.29 2.20 2.11
 4.41 4.17 3.92 3.66 3.49 3.38 3.23 3.14 3.08 2.92 2.78

(Continued)

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TABLE E

F critical values (continued)

Degrees of freedom in the numerator

p 1 2 3 4 5 6 7 8 9

.100 2.89 2.50 2.29 2.16 2.06 2.00 1.94 1.90 1.87
 .050 4.20 3.34 2.95 2.71 2.56 2.45 2.36 2.29 2.24
 28 .025 5.61 4.22 3.63 3.29 3.06 2.90 2.78 2.69 2.61
 .010 7.64 5.45 4.57 4.07 3.75 3.53 3.36 3.23 3.12
 .001 13.50 8.93 7.19 6.25 5.66 5.24 4.93 4.69 4.50
 .100 2.89 2.50 2.28 2.15 2.06 1.99 1.93 1.89 1.86
 .050 4.18 3.33 2.93 2.70 2.55 2.43 2.35 2.28 2.22
 29 .025 5.59 4.20 3.61 3.27 3.04 2.88 2.76 2.67 2.59
 .010 7.60 5.42 4.54 4.04 3.73 3.50 3.33 3.20 3.09
 .001 13.39 8.85 7.12 6.19 5.59 5.18 4.87 4.64 4.45
 .100 2.88 2.49 2.28 2.14 2.05 1.98 1.93 1.88 1.85
 .050 4.17 3.32 2.92 2.69 2.53 2.42 2.33 2.27 2.21
 30 .025 5.57 4.18 3.59 3.25 3.03 2.87 2.75 2.65 2.57
 .010 7.56 5.39 4.51 4.02 3.70 3.47 3.30 3.17 3.07
 .001 13.29 8.77 7.05 6.12 5.53 5.12 4.82 4.58 4.39
 .100 2.84 2.44 2.23 2.09 2.00 1.93 1.87 1.83 1.79
 .050 4.08 3.23 2.84 2.61 2.45 2.34 2.25 2.18 2.12
 40 .025 5.42 4.05 3.46 3.13 2.90 2.74 2.62 2.53 2.45

Degrees of freedom in the denominator

.010 7.31 5.18 4.31 3.83 3.51 3.29 3.12 2.99 2.89
 .001 12.61 8.25 6.59 5.70 5.13 4.73 4.44 4.21 4.02
 .100 2.81 2.41 2.20 2.06 1.97 1.90 1.84 1.80 1.76
 .050 4.03 3.18 2.79 2.56 2.40 2.29 2.20 2.13 2.07
 50 .025 5.34 3.97 3.39 3.05 2.83 2.67 2.55 2.46 2.38
 .010 7.17 5.06 4.20 3.72 3.41 3.19 3.02 2.89 2.78
 .001 12.22 7.96 6.34 5.46 4.90 4.51 4.22 4.00 3.82
 .100 2.79 2.39 2.18 2.04 1.95 1.87 1.82 1.77 1.74
 .050 4.00 3.15 2.76 2.53 2.37 2.25 2.17 2.10 2.04
 60 .025 5.29 3.93 3.34 3.01 2.79 2.63 2.51 2.41 2.33
 .010 7.08 4.98 4.13 3.65 3.34 3.12 2.95 2.82 2.72
 .001 11.97 7.77 6.17 5.31 4.76 4.37 4.09 3.86 3.69
 .100 2.76 2.36 2.14 2.00 1.91 1.83 1.78 1.73 1.69
 .050 3.94 3.09 2.70 2.46 2.31 2.19 2.10 2.03 1.97
 100 .025 5.18 3.83 3.25 2.92 2.70 2.54 2.42 2.32 2.24
 .010 6.90 4.82 3.98 3.51 3.21 2.99 2.82 2.69 2.59
 .001 11.50 7.41 5.86 5.02 4.48 4.11 3.83 3.61 3.44
 .100 2.73 2.33 2.11 1.97 1.88 1.80 1.75 1.70 1.66
 .050 3.89 3.04 2.65 2.42 2.26 2.14 2.06 1.98 1.93
 200 .025 5.10 3.76 3.18 2.85 2.63 2.47 2.35 2.26 2.18
 .010 6.76 4.71 3.88 3.41 3.11 2.89 2.73 2.60 2.50

.001 11.15 7.15 5.63 4.81 4.29 3.92 3.65 3.43 3.26
 .100 2.71 2.31 2.09 1.95 1.85 1.78 1.72 1.68 1.64
 .050 3.85 3.00 2.61 2.38 2.22 2.11 2.02 1.95 1.89
 1000 .025 5.04 3.70 3.13 2.80 2.58 2.42 2.30 2.20 2.13
 .010 6.66 4.63 3.80 3.34 3.04 2.82 2.66 2.53 2.43
 .001 10.89 6.96 5.46 4.65 4.14 3.78 3.51 3.30 3.13

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Tables T-19

TABLE E

F critical values (continued)

Degrees of freedom in the numerator

10 12 15 20 25 30 40 50 60 120 1000

1.84 1.79 1.74 1.69 1.65 1.63 1.59 1.57 1.56 1.52 1.48
 2.19 2.12 2.04 1.96 1.91 1.87 1.82 1.79 1.77 1.71 1.66
 2.55 2.45 2.34 2.23 2.16 2.11 2.05 2.01 1.98 1.91 1.84
 3.03 2.90 2.75 2.60 2.51 2.44 2.35 2.30 2.26 2.17 2.08
 4.35 4.11 3.86 3.60 3.43 3.32 3.18 3.09 3.02 2.86 2.72

1.83 1.78 1.73 1.68 1.64 1.62 1.58 1.56 1.55 1.51 1.47
 2.18 2.10 2.03 1.94 1.89 1.85 1.81 1.77 1.75 1.70 1.65
 2.53 2.43 2.32 2.21 2.14 2.09 2.03 1.99 1.96 1.89 1.82
 3.00 2.87 2.73 2.57 2.48 2.41 2.33 2.27 2.23 2.14 2.05
 4.29 4.05 3.80 3.54 3.38 3.27 3.12 3.03 2.97 2.81 2.66

1.82 1.77 1.72 1.67 1.63 1.61 1.57 1.55 1.54 1.50 1.46
 2.16 2.09 2.01 1.93 1.88 1.84 1.79 1.76 1.74 1.68 1.63
 2.51 2.41 2.31 2.20 2.12 2.07 2.01 1.97 1.94 1.87 1.80
 2.98 2.84 2.70 2.55 2.45 2.39 2.30 2.25 2.21 2.11 2.02
 4.24 4.00 3.75 3.49 3.33 3.22 3.07 2.98 2.92 2.76 2.61

1.76 1.71 1.66 1.61 1.57 1.54 1.51 1.48 1.47 1.42 1.38
 2.08 2.00 1.92 1.84 1.78 1.74 1.69 1.66 1.64 1.58 1.52
 2.39 2.29 2.18 2.07 1.99 1.94 1.88 1.83 1.80 1.72 1.65
 2.80 2.66 2.52 2.37 2.27 2.20 2.11 2.06 2.02 1.92 1.82
 3.87 3.64 3.40 3.14 2.98 2.87 2.73 2.64 2.57 2.41 2.25

1.73 1.68 1.63 1.57 1.53 1.50 1.46 1.44 1.42 1.38 1.33
 2.03 1.95 1.87 1.78 1.73 1.69 1.63 1.60 1.58 1.51 1.45
 2.32 2.22 2.11 1.99 1.92 1.87 1.80 1.75 1.72 1.64 1.56
 2.70 2.56 2.42 2.27 2.17 2.10 2.01 1.95 1.91 1.80 1.70
 3.67 3.44 3.20 2.95 2.79 2.68 2.53 2.44 2.38 2.21 2.05

1.71 1.66 1.60 1.54 1.50 1.48 1.44 1.41 1.40 1.35 1.30

1.99 1.92 1.84 1.75 1.69 1.65 1.59 1.56 1.53 1.47 1.40
2.27 2.17 2.06 1.94 1.87 1.82 1.74 1.70 1.67 1.58 1.49
2.63 2.50 2.35 2.20 2.10 2.03 1.94 1.88 1.84 1.73 1.62
3.54 3.32 3.08 2.83 2.67 2.55 2.41 2.32 2.25 2.08 1.92
1.66 1.61 1.56 1.49 1.45 1.42 1.38 1.35 1.34 1.28 1.22
1.93 1.85 1.77 1.68 1.62 1.57 1.52 1.48 1.45 1.38 1.30
2.18 2.08 1.97 1.85 1.77 1.71 1.64 1.59 1.56 1.46 1.36
2.50 2.37 2.22 2.07 1.97 1.89 1.80 1.74 1.69 1.57 1.45
3.30 3.07 2.84 2.59 2.43 2.32 2.17 2.08 2.01 1.83 1.64
1.63 1.58 1.52 1.46 1.41 1.38 1.34 1.31 1.29 1.23 1.16
1.88 1.80 1.72 1.62 1.56 1.52 1.46 1.41 1.39 1.30 1.21
2.11 2.01 1.90 1.78 1.70 1.64 1.56 1.51 1.47 1.37 1.25
2.41 2.27 2.13 1.97 1.87 1.79 1.69 1.63 1.58 1.45 1.30
3.12 2.90 2.67 2.42 2.26 2.15 2.00 1.90 1.83 1.64 1.43
1.61 1.55 1.49 1.43 1.38 1.35 1.30 1.27 1.25 1.18 1.08
1.84 1.76 1.68 1.58 1.52 1.47 1.41 1.36 1.33 1.24 1.11
2.06 1.96 1.85 1.72 1.64 1.58 1.50 1.45 1.41 1.29 1.13
2.34 2.20 2.06 1.90 1.79 1.72 1.61 1.54 1.50 1.35 1.16
2.99 2.77 2.54 2.30 2.14 2.02 1.87 1.77 1.69 1.49 1.22